

Data Flow Diagram

Date	24 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

The farmer enters soil nutrient values and environmental parameters into the OptiCrop system. The input data is validated before being sent to the crop prediction process. The prediction process uses the agricultural dataset and trained Random Forest model to determine the most suitable crop. The generated recommendation, confidence score, and alternative crop suggestions are stored and displayed back to the farmer for decision-making.

DFD Symbol Legend

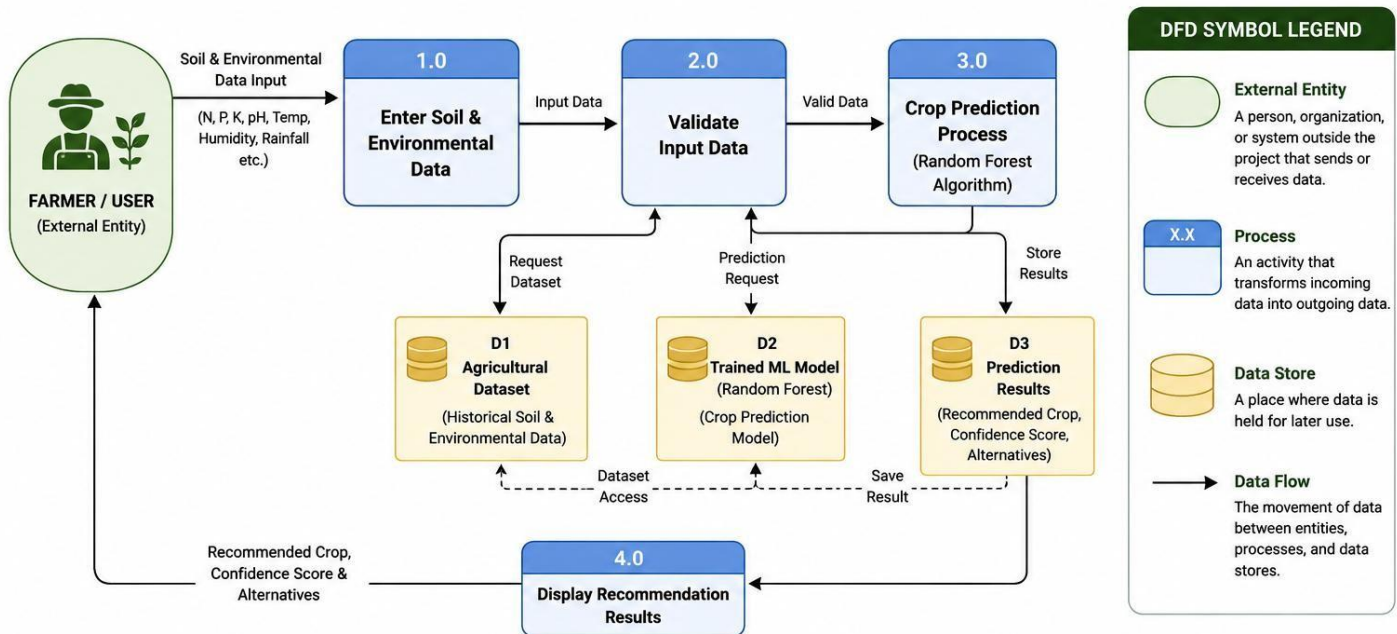
Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).

Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.



DATA FLOW DIAGRAM – OPTICROP

Level 0 DFD – Crop Recommendation System



DESCRIPTION

The farmer provides soil and environmental data to the system. The data is validated and processed using a trained machine learning model (Random Forest) along with historical datasets to generate the best crop recommendation. The results are stored and displayed back to the farmer for better decision making.



BENEFITS

- Accurate Crop Recommendation
- Improved Productivity
- Efficient Resource Utilization
- Informed Decision Making